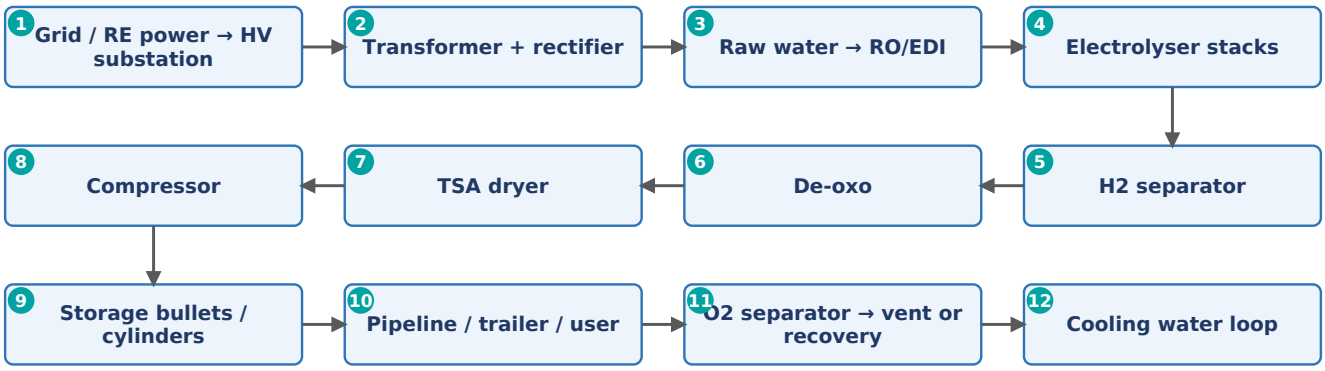
MODULE 5

Industrial Plant Layout & Equipment

PFD · layout · utilities · instrumentation · equipment explorer

5.1 Process flow diagram (green hydrogen plant)



Blocks 11 and 12 run in parallel with the main H_2 path.

5.2 Typical plant layout

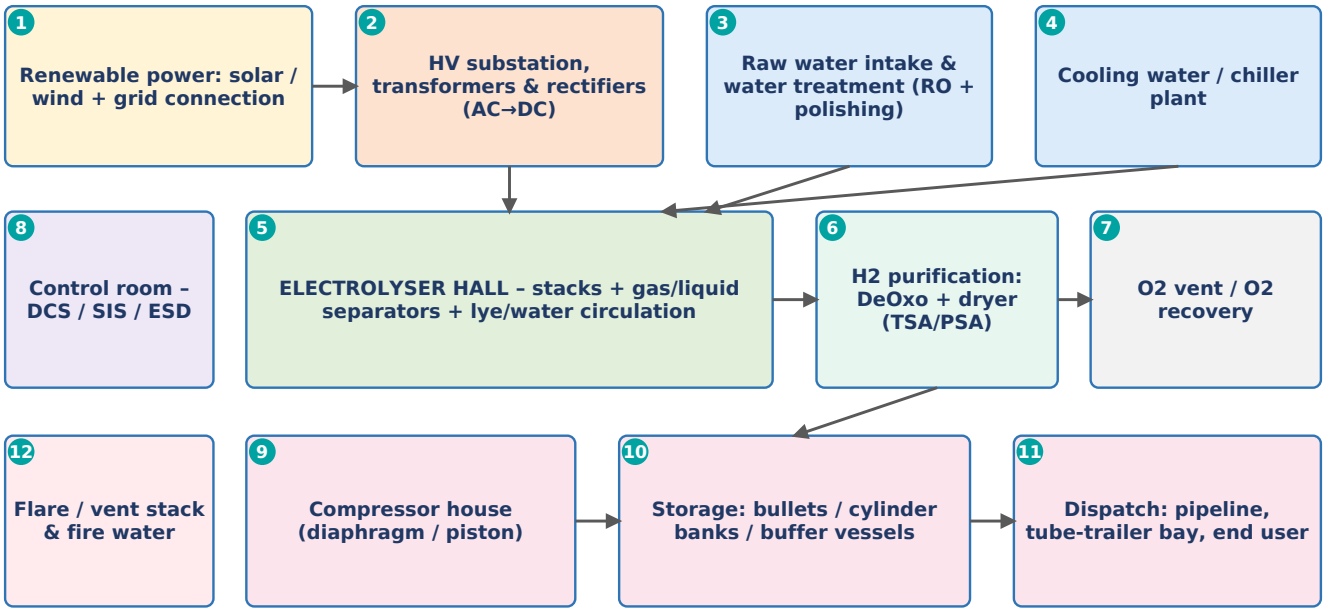


Figure 5.1 – Indicative block layout (not to scale). Hazardous gas areas (5, 6, 9, 10, 11) are separated from the control room (8) and electrical buildings (2); actual separation distances come from the site hazard study and the applicable codes (Module 10).

5.3 Utilities and infrastructure

Utility	Role	Design notes
Electrical	Main power for stacks + BoP	HV substation, transformers, rectifiers, MCCs, UPS for control, earthing & lightning protection
Water	Feed for electrolysis	Raw water storage, RO + EDI/polishing, demin tank; plan for 18–24 kg/kg H_2 [R1]
Cooling	Removes stack & compressor heat	Closed loop with plate HX; cooling tower or dry coolers

Utility	Role	Design notes
Nitrogen	Purging & inerting	Required for start-up, shutdown, maintenance purges
Instrument air	Valve actuation	Dry, oil-free
Fire water	Fire protection & cooling of vessels	Hydrants, deluge on storage per hazard study

5.4 Instrumentation & control

- **DCS / PLC** – production control: stack current, temperatures, pressures, levels, flow.
- **Safety Instrumented System (SIS)** – independent trips: H_2-in-O_2 high, pressure high/low, separator level, gas detection, ESD push-buttons.
- **Analysers** – O_2 in H_2, H_2 in O_2, dew point, water conductivity.
- **Fire & gas system** – H_2 detectors at high points, UV/IR flame detectors (H_2 flames are hard to see).

5.5 Equipment explorer

Each card gives: working principle · main components · plant function · operating parameters · maintenance points.

Equipment: Transformer-rectifier	Details
Working principle	Steps down voltage and converts AC to DC with thyristor or IGBT bridges
Main components	Transformer, rectifier bridge, filters, DC busbars
Function in plant	Supplies controlled DC to each stack
Operating parameters	Output current/voltage per stack rating
Maintenance	Thermal imaging of connections; cooling fans/water; harmonic checks

Equipment: Reverse osmosis + EDI	Details
Working principle	Pressure forces water through semi-permeable membranes; EDI polishes with ion-exchange + electric field
Main components	Pre-filters, high-pressure pump, RO vessels, EDI modules
Function in plant	Makes demin water
Operating parameters	Conductivity at outlet to OEM spec
Maintenance	Membrane cleaning (CIP), antiscalant dosing, filter changes

Equipment: Electrolyser stack	Details
Working principle	Electrochemical water splitting (Module 4)
Main components	Cells, bipolar plates, seals, end plates
Function in plant	Produces H_2 and O_2
Operating parameters	Temperature, pressure, current density per type [R1]
Maintenance	Cell-voltage trending (degradation), re-torque, stack refurbishment

Equipment: Gas-liquid separator	Details
Working principle	Gravity + demister separate liquid from gas
Main components	Vessel, demister pad, level transmitters
Function in plant	Dries gas, returns liquid
Operating parameters	Level & pressure balanced H_2/O_2 sides
Maintenance	Level instrument calibration, demister inspection

Equipment: De-oxo reactor	Details
Working principle	Catalytic recombination of O_2 with H_2
Main components	Catalyst bed, heater, O_2 analyser
Function in plant	Removes O_2
Operating parameters	Bed temperature; inlet O_2
Maintenance	Catalyst replacement; analyser calibration

Equipment: TSA dryer	Details
Working principle	Adsorbs water on molecular sieve; regenerated by heating
Main components	Twin beds, heater, cooler, switching valves
Function in plant	Dries H_2
Operating parameters	Outlet dew point
Maintenance	Valve leak tests; sieve replacement

Equipment: PSA unit (SMR plants)	Details
Working principle	Impurities adsorb at high pressure, desorb at low pressure
Main components	Multiple adsorber beds, valve skid, tail-gas drum
Function in plant	Purifies H_2
Operating parameters	Cycle time; recovery vs purity
Maintenance	Valve maintenance; adsorbent life

Equipment: Diaphragm compressor	Details
Working principle	Hydraulically driven metal diaphragm compresses gas – no contact with oil
Main components	Diaphragm set, hydraulic system, valves
Function in plant	Raises pressure, oil-free
Operating parameters	Stages & intercooling per pressure ratio
Maintenance	Diaphragm leak detection & replacement

Equipment: Storage vessel / cylinder bank	Details
Working principle	Holds compressed H_2
Main components	Shell, nozzles, PRVs, instruments
Function in plant	Buffer storage
Operating parameters	Design pressure per code
Maintenance	Periodic inspection, fatigue assessment, PRV testing

Equipment: Gas detector	Details
Working principle	Catalytic, thermal-conductivity or electrochemical sensing
Main components	Sensor head, transmitter
Function in plant	Leak detection → alarm/trip
Operating parameters	Set-points as % of LFL per hazard study
Maintenance	Bump test & calibration

Quiz & revision – Module 5

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Which system trips the plant on high H2-in-O2?
a) DCS **b)** SIS **c)** SCADA historian **d)** Water analyser

Q2. Why are diaphragm compressors common for H2?
a) Cheapest **b)** Oil-free gas path **c)** No cooling needed **d)** Single stage to 900 bar

Q3. Nitrogen in an H2 plant is mainly used for:
a) Cooling **b)** Purging/inerting **c)** Electrolyte **d)** Fuel

Q4. Where should H2 detectors be located indoors?
a) Near floor **b)** At high points / ceiling **c)** Outside only **d)** In drains

Q5. Which unit is typical of SMR plants rather than electrolysis plants?
a) De-oxo **b)** Rectifier **c)** PSA **d)** EDI

Answer key
Q1 – b: independent safety layer; **Q2 – b**: ; **Q3 – b**: ; **Q4 – b**: H2 is buoyant; **Q5 – c**: removes CO2/CO/CH4